

3D genomic analysis reveals novel enhancer-hijacking caused by complex structural alterations that drive oncogene overexpression.

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Cancer genomes are composed of many complex structural alterations on chromosomes and extrachromosomal DNA (ecDNA), making it difficult to identify non-coding enhancer regions that are hijacked to activate oncogene expression. Here, we describe a 3D genomics-based analysis called HAPI (Highly Active Promoter Interactions) to characterize enhancer hijacking. HAPI analysis of HiChIP data from 34 cancer cell lines identified enhancer hijacking events that activate both known and potentially novel oncogenes such as MYC, CCND1, ETV1, CRKL, and ID4. Furthermore, we found enhancer hijacking among multiple oncogenes from different chromosomes, often including MYC, on the same complex amplicons such as ecDNA. We characterized a MYC - ERBB2 chimeric ecDNA, in which ERBB2 heavily hijacks MYC's enhancers. Notably, CRISPRi of the MYC promoter led to increased interaction of ERBB2 with MYC enhancers and elevated ERBB2 expression. Our HAPI analysis tool provides a robust strategy to detect enhancer hijacking and reveals novel insights into oncogene activation.

Structural variation cooperates with permissive chromatin to control enhancer hijacking-mediated oncogenic transcription.

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Structural variants (SVs) involving enhancer hijacking can rewire chromatin topologies to cause oncogene activation in human cancers, including hematologic malignancies; however, because of the lack of tools to assess their effects on gene regulation and chromatin organization, the molecular determinants for the functional output of enhancer hijacking remain poorly understood. Here, we developed a multimodal approach to integrate genome sequencing, chromosome conformation, chromatin state, and transcriptomic alteration for quantitative analysis of transcriptional effects and structural reorganization imposed by SVs in leukemic genomes. We identified known and new pathogenic SVs, including recurrent t(5;14) translocations that cause the hijacking of BCL11B enhancers for the allele-specific activation of TLX3 in a subtype of pediatric leukemia. Epigenetic perturbation of SV-hijacked BCL11B enhancers impairs TLX3 transcription, which are required for the growth of t(5;14) leukemia cells. By CRISPR engineering of patient-derived t(5;14) in isogenic leukemia cells, we uncovered a new mechanism whereby the transcriptional output of SV-induced BCL11B enhancer hijacking is dependent on the loss of DNA hypermethylation at the TLX3 promoter. Our results highlight the importance of the cooperation between genetic alteration and permissive chromatin as a critical determinant of SV-mediated oncogene activation, with implications for understanding aberrant gene transcription after epigenetic therapies in patients with leukemia. Hence, leveraging the interdependency of genetic alteration on chromatin variation may provide new opportunities to reprogram gene regulation as targeted interventions in human disease.

Transcriptional dysregulation by aberrant enhancer activation and rewiring in cancer.

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Cell identity is controlled by regulatory elements, such as promoters, enhancers, and insulators, within

the genome. These regulatory elements interact in the nucleus and form tissue-specific chromatin structures. Dysregulation of these elements and their interactions can lead to loss of cell identity and promote the development of diseases such as cancer. Tumor cells acquire aberrantly activated enhancers at oncogenic driver genes through various mechanisms. Small genomic changes such as mutations, insertions, and amplifications can form aberrant enhancers. Genomic rearrangements at the chromosomal level, including translocations and inversions, are also often observed in cancers. These rearrangements can result in repositioning of enhancers to locations near tumor-type-specific oncogenes. Chromatin structural changes caused by genomic or epigenomic changes lead to mis-interaction between enhancers and proto-oncogenes, ultimately contributing to tumorigenesis through activation of oncogenic signals. Additional epigenomic mechanisms can also cause aberrant enhancer activation, including those associated with overexpression of oncogenic transcription factors and the mutation of transcriptional cofactors. Exogenous viral DNA can also lead to enhancer aberrations. Here, we review the mechanisms underlying aberrant oncogene activation through enhancer activation and rewiring, both of which are caused by genomic or epigenomic alterations in non-coding regions.

Modelling and drug targeting of a myeloid neoplasm with atypical 3q26/MECOM rearrangement using patient-specific iPSCs.

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Structural variations involving enhancer hijacking induce aberrant oncogene expression and cause tumorigenesis. A rare translocation, t(3;8)(q26.2;q24), is associated with MECOM and MYC rearrangement, causing myeloid neoplasms with a dismal prognosis. The most recent World Health Organization classification recognises myeloid neoplasms with MECOM rearrangement as acute myeloid leukaemia (AML) with defining genetic abnormalities. Recently, the increasing use of induced pluripotent stem cell (iPSC) technology has helped elucidate the pathogenic processes of haematological malignancies. However, its utility for investigating enhancer hijacking in myeloid neoplasms remains unclear. In this study, we generated iPSC lines from patients with myelodysplastic syndromes (MDS) harbouring t(3;8)(q26.2;q24) and differentiated them into haematopoietic progenitor cells to model the pathophysiology of MDS with t(3;8)(q26.2;q24). Our iPSC model reproduced the primary patient's MECOM expression changes and histone H3 lysine 27 acetylation (H3K27ac) patterns in the MECOM promoter and MYC blood enhancer cluster (BENC). Furthermore, we revealed the apoptotic effects of the bromodomain and extra-terminal motif (BET) inhibitor on iPSC-derived MDS cells by suppressing activated MECOM. Our study demonstrates the usefulness of iPSC models for uncovering the precise mechanism of enhancer hijacking due to chromosomal structural changes and discovering potential therapeutic drug candidates for cancer treatment.

Spatial and clonality-resolved 3D cancer genome alterations reveal enhancer-hijacking as a potential prognostic marker for colorectal cancer.

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The regulatory effect of non-coding large-scale structural variations (SVs) on proto-oncogene activation remains unclear. This study investigated SV-mediated gene dysregulation by profiling 3D cancer genome maps from 40 patients with colorectal cancer (CRC). We developed a machine learning-based method for

spatial characterization of the altered 3D cancer genome. This revealed a frequent establishment of "de novo chromatin contacts" that can span multiple topologically associating domains (TADs) in addition to the canonical TAD fusion/shuffle model. Using this information, we precisely identified super-enhancer (SE)-hijacking and its clonal characteristics. Clonal SE-hijacking genes, such as TOP2B, are recurrently associated with cell-cycle/DNA-processing functions, which can potentially be used as CRC prognostic markers. Oncogene activation and increased drug resistance due to SE-hijacking were validated by reconstructing the patient's SV using CRISPR-Cas9. Collectively, the spatial and clonality-resolved analysis of the 3D cancer genome reveals regulatory principles of large-scale SVs in oncogene activation and their clinical implications.

Deregulation of enhancer structure, function, and dynamics in acute lymphoblastic leukemia.

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Enhancers control dynamic changes in gene expression and orchestrate the tightly controlled transcriptional circuitries that direct and coordinate cell growth, proliferation, survival, lineage commitment, and differentiation during lymphoid development. Enhancer hijacking and neoenhancer formation at oncogene loci, as well as aberrant activation of oncogene-associated enhancers, can induce constitutive activation of self-perpetuating oncogenic transcriptional circuitries, and contribute to the malignant transformation of immature lymphoid progenitors in acute lymphoblastic leukemia (ALL). In this review, we present recent discoveries of the role of enhancer dynamics in mouse and human lymphoid development, and discuss how genetic and epigenetic alterations of enhancer function can promote leukemogenesis, and potential strategies for targeting the enhancer machinery in the treatment of ALL.

Enhancer Hijacking Drives Oncogenic BCL11B Expression in Lineage-Ambiguous Stem Cell Leukemia.

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Lineage-ambiguous leukemias are high-risk malignancies of poorly understood genetic basis. Here, we describe a distinct subgroup of acute leukemia with expression of myeloid, T lymphoid, and stem cell markers driven by aberrant allele-specific deregulation of BCL11B, a master transcription factor responsible for thymic T-lineage commitment and specification. Mechanistically, this deregulation was driven by chromosomal rearrangements that juxtapose BCL11B to superenhancers active in hematopoietic progenitors, or focal amplifications that generate a superenhancer from a noncoding element distal to BCL11B. Chromatin conformation analyses demonstrated long-range interactions of rearranged enhancers with the expressed BCL11B allele and association of BCL11B with activated hematopoietic progenitor cell cis-regulatory elements, suggesting BCL11B is aberrantly co-opted into a gene regulatory network that drives transformation by maintaining a progenitor state. These data support a role for ectopic BCL11B expression in primitive hematopoietic cells mediated by enhancer hijacking as an oncogenic driver of human lineage-ambiguous leukemia. SIGNIFICANCE: Lineage-ambiguous leukemias pose significant diagnostic and therapeutic challenges due to a poorly understood molecular and cellular basis. We identify oncogenic deregulation of BCL11B driven by diverse

structural alterations, including de novo superenhancer generation, as the driving feature of a subset of lineage-ambiguous leukemias that transcend current diagnostic boundaries. This article is highlighted in the In This Issue feature, p. 2659.

Revealing the impact of structural variants in multiple myeloma.

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The landscape of structural variants (SVs) in multiple myeloma remains poorly understood. Here, we performed comprehensive analysis of SVs in a large cohort of 752 multiple myeloma patients by low coverage long-insert whole genome sequencing. We identified 68 SV hotspots involving 17 new candidate driver genes, including the therapeutic targets BCMA (TNFRSF17), SLAMF and MCL1. Catastrophic complex rearrangements termed chromothripsis were present in 24% of patients and independently associated with poor clinical outcomes. Templated insertions were the second most frequent complex event (19%), mostly involved in super-enhancer hijacking and activation of oncogenes such as CCND1 and MYC. Importantly, in 31% of patients two or more seemingly independent putative driver events were caused by a single structural event, demonstrating that the complex genomic landscape of multiple myeloma can be acquired through few key events during tumor evolutionary history. Overall, this study reveals the critical role of SVs in multiple myeloma pathogenesis.

Prediction of the 3D cancer genome from whole-genome sequencing using InfoHiC.

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The 3D genome prediction in cancer is crucial for uncovering the impact of structural variations (SVs) on tumorigenesis, especially when they are present in noncoding regions. We present InfoHiC, a systemic framework for predicting the 3D cancer genome directly from whole-genome sequencing (WGS). InfoHiC utilizes contig-specific copy number encoding on the SV contig assembly, and performs a contig-to-total Hi-C conversion for the cancer Hi-C prediction from multiple SV contigs. We showed that InfoHiC can predict 3D genome folding from all types of SVs using breast cancer cell line data. We applied it to WGS data of patients with breast cancer and pediatric patients with medulloblastoma, and identified neo topologically associating domains. For breast cancer, we discovered super-enhancer hijacking events associated with oncogenic overexpression and poor survival outcomes. For medulloblastoma, we found SVs in noncoding regions that caused super-enhancer hijacking events of medulloblastoma driver genes (GFI1, GFI1B, and PRDM6). In addition, we provide trained models for cancer Hi-C prediction from WGS at <https://github.com/dmcb-gist/InfoHiC>, uncovering the impacts of SVs in cancer patients and revealing novel therapeutic targets.

Genomic landscape of Down syndrome-associated acute lymphoblastic leukemia.

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Trisomy 21, the genetic cause of Down syndrome (DS), is the most common congenital chromosomal anomaly. It is associated with a 20-fold increased risk of acute lymphoblastic leukemia (ALL) during

childhood and results in distinctive leukemia biology. To comprehensively define the genomic landscape of DS-ALL, we performed whole-genome sequencing and whole-transcriptome sequencing (RNA-Seq) on 295 cases. Our integrated genomic analyses identified 15 molecular subtypes of DS-ALL, with marked enrichment of CRLF2-r, IGH::IGF2BP1, and C/EBP altered (C/EBPalt) subtypes compared with 2257 non-DS-ALL cases. We observed abnormal activation of the CEBPD, CEBPA, and CEBPE genes in 10.5% of DS-ALL cases via a variety of genomic mechanisms, including chromosomal rearrangements and noncoding mutations leading to enhancer hijacking. A total of 42.3% of C/EBP-activated DS-ALL also have concomitant FLT3 point mutations or insertions/deletions, compared with 4.1% in other subtypes. CEBPD overexpression enhanced the differentiation of mouse hematopoietic progenitor cells into pro-B cells in vitro, particularly in a DS genetic background. Notably, recombination-activating gene-mediated somatic genomic abnormalities were common in DS-ALL, accounting for a median of 27.5% of structural alterations, compared with 7.7% in non-DS-ALL. Unsupervised hierarchical clustering analyses of CRLF2-rearranged DS-ALL identified substantial heterogeneity within this group, with the BCR::ABL1-like subset linked to an inferior event-free survival, even after adjusting for known clinical risk factors. These results provide important insights into the biology of DS-ALL and point to opportunities for targeted therapy and treatment individualization.
